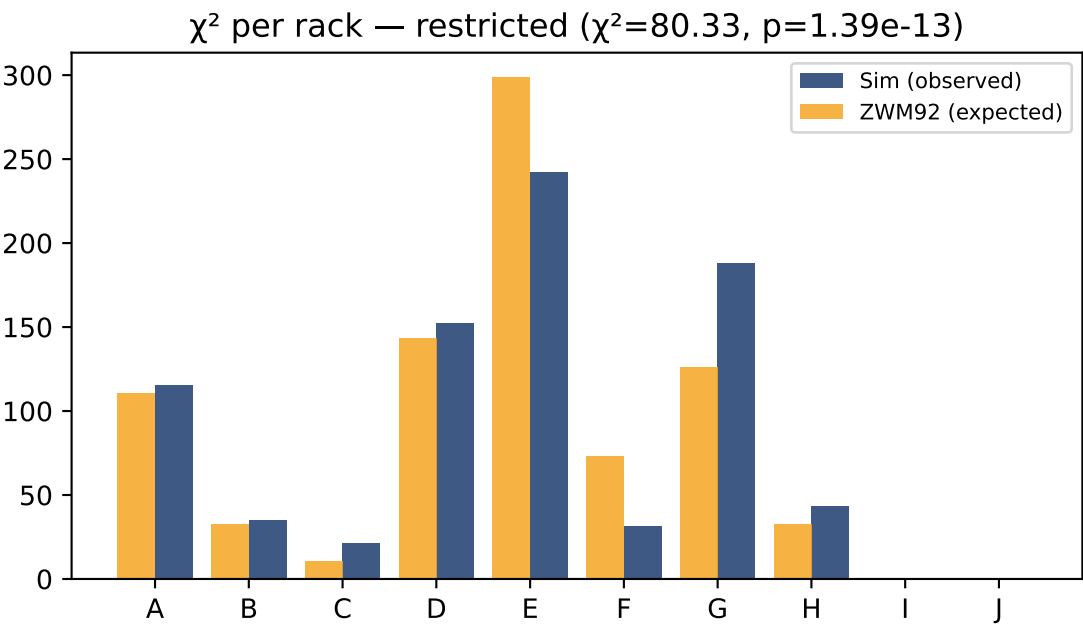


V&V5 — Operational Validity

χ^2 + paired-t + Holm-Wilcoxon



Paired t (log daily picks per material):
 $t = -8.520$, $p = 8.37e-17$, $n = 770$

Restricted $\chi^2 = 80.3$ (Cramer's $V = 0.104$ — small effect); pooled (Cochran OK): $\chi^2 = 80.$

Replication volume calibration: PASS — Baseline (Actual SAP): simulated arrivals/day within 8.8% of ZW

Unpooled Cochran note: Cochran's rule violated: 2 of 10 cells have expected count < 5 — [('I', 0.01), ('J', 0.04)].
Unpooled p-value

CRN paired-by-replication t-tests vs Heuristic (N=20)

